

Experiment-14

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Course Code: -MLA0305

Title: Sampling-Based Planning Using a Learned Environment Model

Aim:

To implement sampling-based planning techniques for generating optimal action sequences using a learned environment model.

Procedure:

1. Develop or obtain an approximate environment model.
2. Observe the current state.
3. Generate multiple candidate action sequences.
4. Simulate each sequence using the learned model.
5. Calculate the expected cumulative reward for each sequence.
6. Select the sequence with the highest predicted reward.
7. Execute the first action of the selected sequence.
8. Observe the new state.
9. Repeat the planning process.
10. Compare predicted and actual rewards.



The screenshot shows a Jupyter Notebook interface. The top menu bar includes File, Edit, View, Insert, Runtime, Tools, and Help. Below the menu is a toolbar with icons for Commands, Code, Text, and Run all. The main area displays a code cell with the following Python code:

```
[6] import numpy as np
rng=np.random.default_rng(4); means={'A':5,'B':9,'C':7}; scores={a:rng.normal(m,2,200).mean() for a,m in means.items()}
print(scores); print("Best action:",max(scores,key=scores.get))
```

The output of the code cell is:

```
{'A': np.float64(5.09582148206032), 'B': np.float64(9.040719575520615), 'C': np.float64(6.931239125384791)}
Best action: B
```

Result:

Sampling-based planning successfully generated action sequences using the learned model. The agent selected actions based on predicted future rewards.